

# Week 1 Progress Report

## 1. Objectives for Week 1:

- Identify and define the project problem statement.
- Conduct an initial literature review on existing ML and DL approaches.
- Shortlist relevant datasets (X-ray and CT scans).

## 2. Activities Completed:

### 1. Problem Identification:

- Finalized the problem statement: “Developing a machine learning-based diagnostic system for early lung cancer detection using X-ray and CT scan images.”
- Documented the significance of early detection and challenges in current methods.

### 2. Literature Review:

- Reviewed 6 key research papers, including works by **Kumar et al. (2020)**, **Zhang et al. (2019)**, and **Wang et al. (2018)**.
- Summarized strengths and limitations of CNN, SVM, Transfer Learning, and Random Forest methods.

### 3. Dataset Exploration:

- Identified **NIH Chest X-ray Dataset** and **LIDC-IDRI CT Dataset**.
- Analyzed dataset size, image formats, and labeling methods.

## 3. Challenges Faced:

- Difficulty in finding combined X-ray + CT datasets.
- Some research papers required paid access.

## 4. Outcomes of Week 1:

- Finalized problem statement and objectives.
- Created a literature summary matrix comparing models and their accuracy.
- Prepared a list of datasets to be downloaded in Week 2.

# Week 2 Progress Report

## 1. Objectives for Week 2:

- Download and organize datasets.
- Begin preprocessing of images.
- Design system architecture draft.

## 2. Activities Completed:

### 1. Dataset Preparation:

- Downloaded **10,000+ chest X-ray images** from NIH dataset.
- Downloaded **1,500 CT scan images** from LIDC-IDRI dataset.
- Organized data into cancerous and non-cancerous folders.

### 2. Preprocessing Pipeline:

- Implemented noise removal using Gaussian filters.
- Standardized image sizes to 224x224 pixels.
- Applied augmentation (rotation, flipping, scaling) to balance classes.

### 3. System Architecture Draft:

- Designed a 3-layer architecture:
  - **Frontend:** React-based UI.
  - **Backend:** Flask API for model inference.
  - **Model Layer:** ML models (CNN, SVM, Random Forest).

## 3. Challenges Faced:

- Preprocessing CT scans required more computational resources than expected.
- Augmentation increased dataset size, leading to storage issues.

## 4. Outcomes of Week 2:

- Fully prepared dataset (cleaned, balanced, and ready for model training).
- Preprocessing pipeline documented for reproducibility.
- Initial system architecture drafted for review.